

Power up your Fabric Development with DAX Studio and Tabular Editor

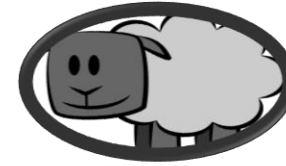


Austin Texas Power BI and Fabric User Group

August 2024

Jason Romans

Senior BI Engineer
Builder of Models



The Dax Shepherd



Lives in Nashville, Tennessee, United States



Started as SQL Server DBA



Transitioned to the Microsoft BI Stack



Work on everything from SQL Server Integration Services, SQL Server Database, Analysis Services, and Power BI



Simple Talk Author at Redgate



Favorite Data Model

Shoulders of Giants

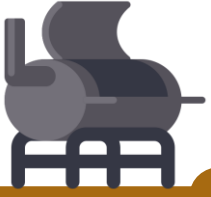


Our Journey



1. Intro
2. Performance Analyzer
3. DAX Studio
4. Tabular Editor
5. Conclusion

Our Journey



1. Intro

2. Performance Analyzer

3. DAX Studio

4. Tabular Editor

5. Conclusion

Power BI is Part of Fabric

- If you have a Fabric capacity
 - Enhances what you can do with Power BI
 - OneLake
 - Warehouse
 - Default Semantic Model
- What we cover applies to Power BI Pro, Power BI Premium, and Fabric

Exam DP-600: Implementing Analytics Solutions Using Microsoft Fabric

Design and build semantic models

- Identify use cases for DAX Studio and **Tabular Editor 2**

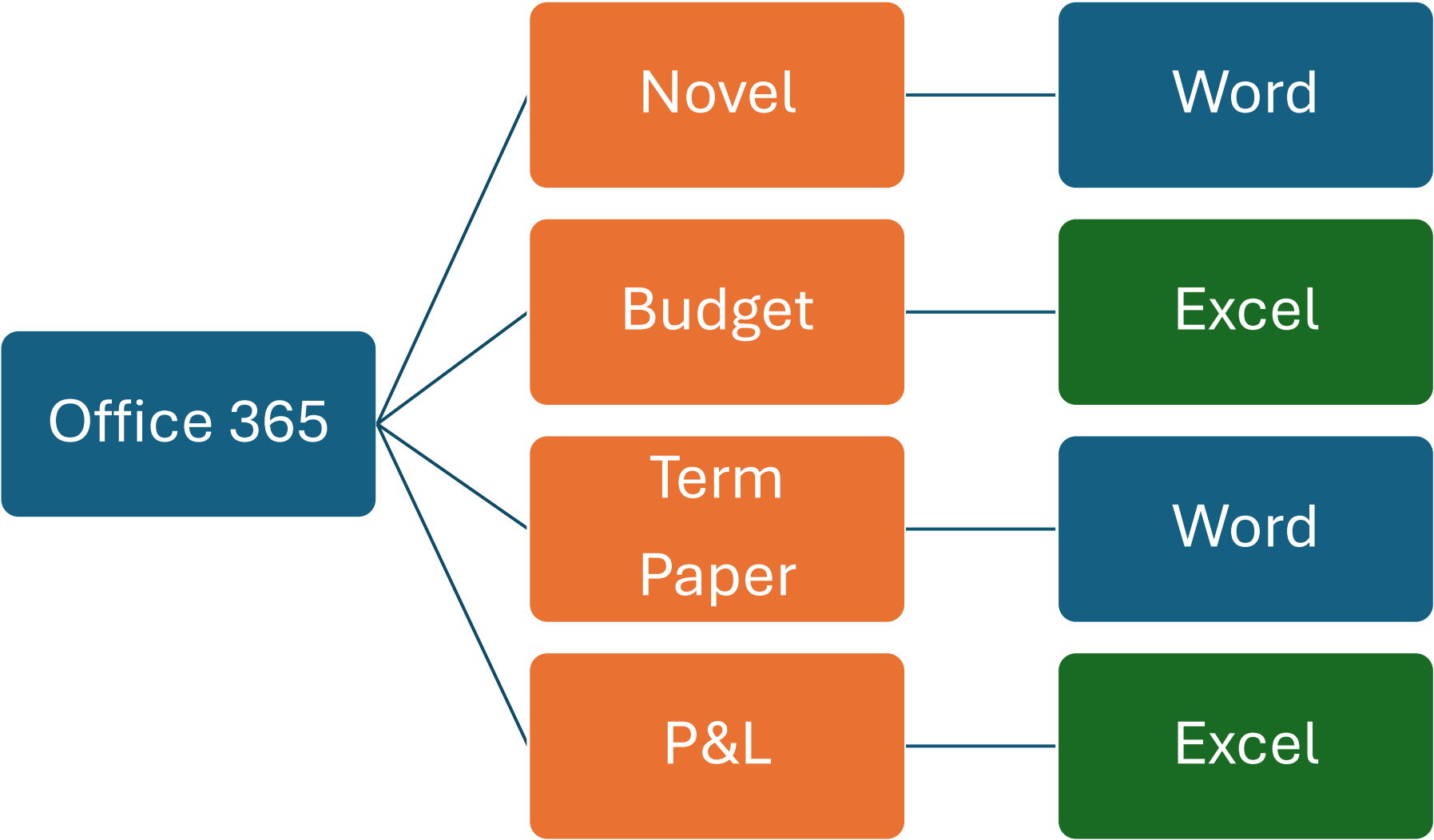
Optimize enterprise-scale semantic models

- Improve DAX performance by using DAX Studio
- Optimize a semantic model by using **Tabular Editor 2**

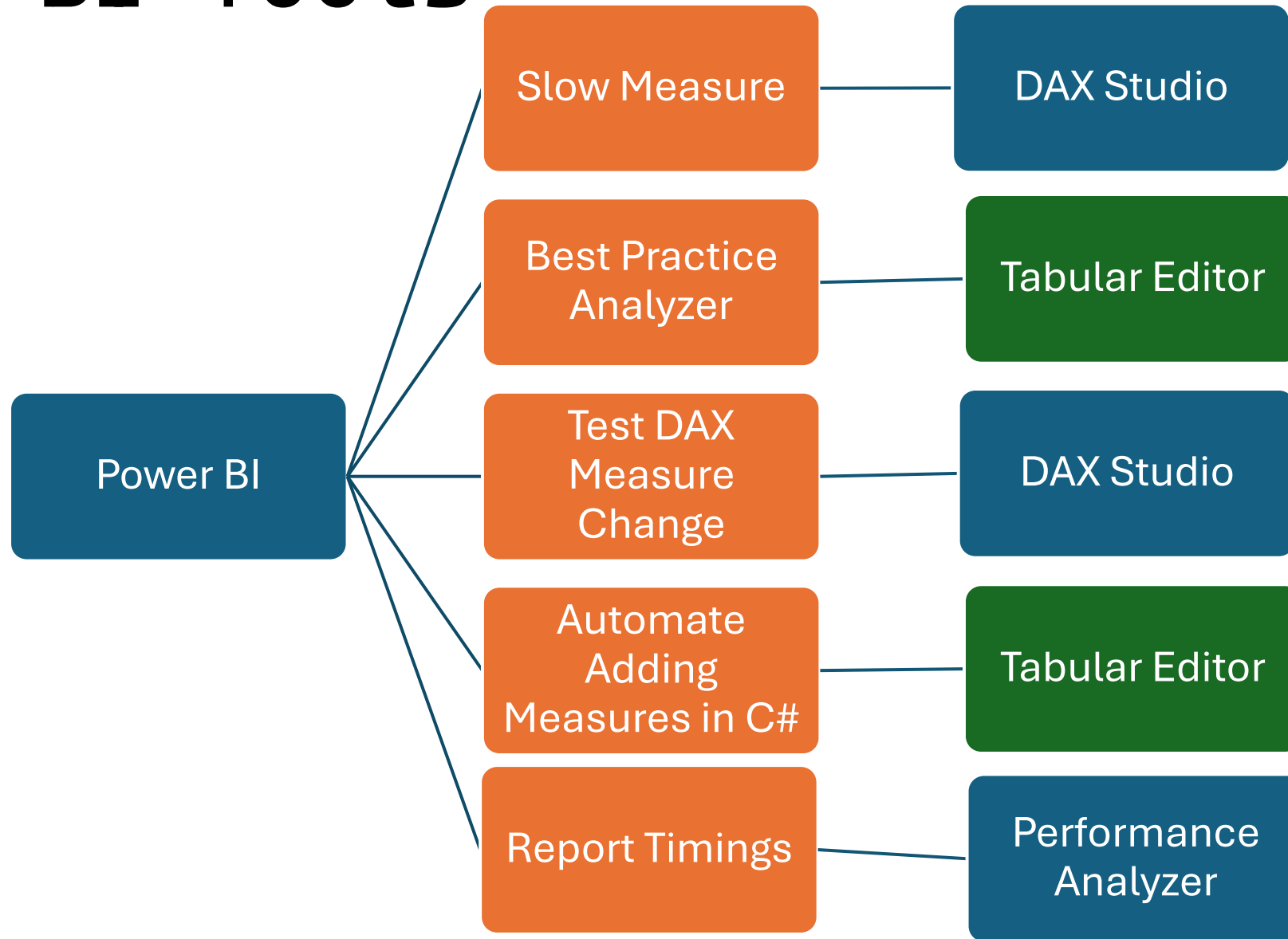
Why is my report slow?

- Motivating Factor
- This can have a lasting impact
 - Power BI Developer Skills
 - Career Development

Office 365



Power BI Tools



Our Journey



1. Intro

2. Performance Analyzer

3. DAX Studio

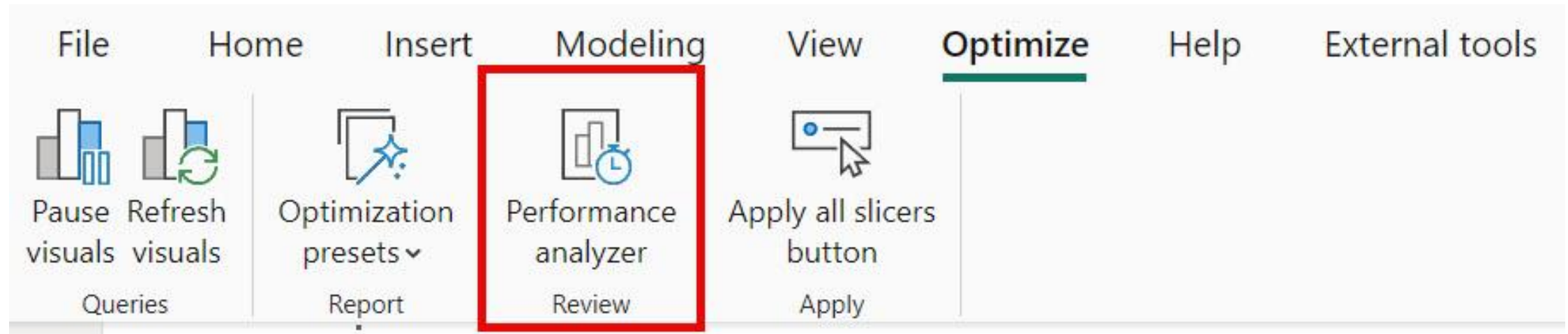
4. Tabular Editor

5. Conclusion

Performance Analyzer

- Not an External Tool
- Built into Power BI Desktop

Performance Analyzer



Performance Analyzer

Start Recording

Filters

🔍 Search

Filters on this page

...

Add data fields here

Filters on all pages

...

Year

is 2020

Add data fields here

Performance analyzer

▶ Start recording

🔄 Refresh visuals

⏹ Stop

Start monitoring your report to see details about the time taken by each visual to query for its data and render the result.

Visualizations

Build visual

<

Performance Analyzer Recording

Performance analyzer

▶ Start recording

↺ Refresh visuals

⏹ Stop

◇ Clear

📄 Export

Name	Duration (ms) ↓
🕒 Recording started (5/1/2...	-
Interact with your report t...	-

Performance Analyzer

Refresh Visuals

Performance analyzer >>

▶ Start recording ↺ Refresh visuals ⌛ Stop

◇ Clear ↗ Export

Name	Duration (ms) ↓
⌚ Recording started (5/1/2...	-
Interact with your report t...	-

Individual Refresh Button

Note: Values for Other will be different

Date	Total Quantity				
Wednesday, January 01, 2020	7,759				
Thursday, January 02, 2020	8,256				
Friday, January 03, 2020	5,482				
Saturday, January 04, 2020	8,608				
Sunday, January 05, 2020	1,144				
Monday, January 06, 2020	3,823				
Tuesday, January 07, 2020	4,414				

List of Visuals

Performance analyzer >>

▶ Start recording ↺ Refresh visuals ⊞ Stop

◇ Clear 📄 Export

Name	Duration (ms) ↓
↺ Refreshed visual	-
⊞ Card	65
⊞ Card	66
⊞ Table	102

Expanded View

Performance analyzer >>

▶ Start recording ↻ Refresh visuals ⏹ Stop

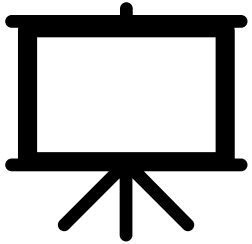
⧫ Clear 📄 Export

Name	Duration (ms) ↓
↻ Refreshed visual	-
☐ Card	65
DAX query	3
Visual display	3
Other	59
📄 Copy query	
📄 Run in DAX Query View	
☐ Card	66
DAX query	3
Visual display	4
Other	59
📄 Copy query	
📄 Run in DAX Query View	
☐ Table	102
DAX query	29
Visual display	31
Other	42
📄 Copy query	
📄 Run in DAX Query View	

Detail on Card

 Refreshed visual	-
 Card	65
DAX query	3
Visual display	3
Other	59
 Copy query	
 Run in DAX Query View	

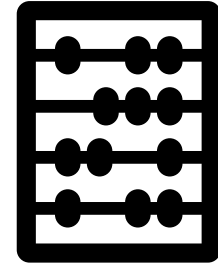
3 Numbers



Visual Display



Other



DAX Query

Visual Display

🔄 Refreshed visual	-
📄 Card	65
DAX query	3
Visual display	3
Other	59
📄 Copy query	
📄 Run in DAX Query View	

- Visual Display
 - Time spent on producing the visual
- Restaurant
 - Plating the food

Slow Visual



Visual Display – Solutions

- Reduce the complexity of visual
 - Granularity of visual
- Use a Background Image
- More in Resources

Other

🔄 Refreshed visual	-
📄 Card	65
DAX query	3
Visual display	3
Other	59
📄 Copy query	
📄 Run in DAX Query View	

- Other
 - Time waiting until DAX Query can be executed
- Restaurant
 - Waiting to order

Other – Solutions

- Reduce Visualizations on the Page
- Evaluate where the bottleneck is
 - May be a combination of other factors

DAX Query

🔄 Refreshed visual	-
📄 Card	65
DAX query	3
Visual display	3
Other	59
📄 Copy query	
📄 Run in DAX Query View	

- DAX Query
 - Time it takes to execute the DAX query
- Restaurant
 - Time to prepare the food

Slow DAX Query

Performance analyzer >>

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms) ↓
Refreshed visual	-
+ Card	52
+ Card	52
- Table	28059
DAX query	27992
Visual display	30
Other	37
Copy query	
Run in DAX Query View	

DAX Query – Solutions



A screenshot of a DAX query performance table. The table has two columns: the first column lists the components of the query, and the second column shows the time spent in milliseconds. The 'DAX query' row is highlighted with a red border. Below the table are two icons: 'Copy query' and 'Run in DAX Query View'.

Refreshed visual	-
Card	65
DAX query	3
Visual display	3
Other	59

Copy query

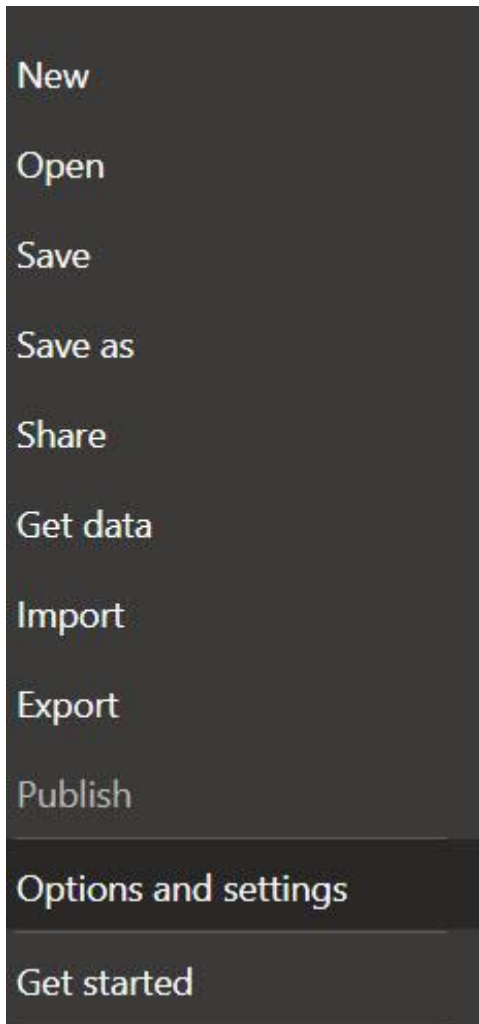
Run in DAX Query View

- Investigate query
 - Where is most of the query time spent
- Can the DAX code be optimized?
 - Rewrite the DAX code

Run in DAX Query View (GA)

	Table	102
	DAX query	29
	Visual display	31
	Other	42
	Copy query	
	Run in DAX Query View	

Enable DAX Query View (Before May 2024 Release)



Options and settings



DAX Query View – Preview Features

Options

GLOBAL

Data Load
Power Query Editor
DirectQuery
R scripting
Python scripting
Security
Privacy
Regional Settings
Updates
Usage Data
Diagnostics
Preview features
Save and Recover
Report settings
Copilot (preview)

CURRENT FILE

Data Load
Regional Settings
Privacy
Auto recovery
Published semantic model settings
Query reduction
Report settings

- ☒ Metrics visual [Learn more](#)
- ☒ Quick measure suggestions [Learn more](#) | [Share feedback](#)
- ☒ Field parameters [Learn more](#)
- ☒ Enhanced row-level security editor [Learn more](#)
- ☐ On-object interaction [Learn more](#) | [Share feedback](#)
- ☐ Power BI Home in Desktop [Learn more](#) | [Share feedback](#)
- ☒ Set sensitivity label on exported PDF [Learn more](#)
- ☒ Dynamic format string for measures [Learn more](#)
- ☒ Save to OneDrive and SharePoint [Learn more](#)
 - ☒ Share to OneDrive and SharePoint [Learn more](#)
- ☐ Enhanced publish dialogs [Learn more](#)
- ☒ Power BI Project (.pbip) save option [Learn more](#)
 - ☒ Store semantic model using TMDL format [Learn more](#)
- ☒ New card visual [Learn more](#)
- ☒ Button slicer visual
- ☐ Less elevated user support [Learn more](#)
- ☒ Model explorer and Calculation group authoring [Learn more](#)
- ☒ Auto-create mobile layout [Learn more](#) | [Share feedback](#)
- ☒ DAX query view [Learn more](#) | [Share feedback](#)
 - ☒ DAX query view with Copilot [Learn more](#) | [Share feedback](#)
- ☒ Summary with Copilot visual [Learn more](#)
- ☐ Improve Q&A with Copilot [Learn more](#)
- ☒ Measure descriptions with Copilot [Learn more](#) | [Share feedback](#)
- ☒ Visual calculations [Learn more](#) | [Share feedback](#)
- ☒ Copilot chat pane in report view [Learn more](#) | [Share feedback](#)

 DAX queries will be saved to your model They won't be visible when published in the Power BI service. [Learn more](#)

Run

```
1 DEFINE
2     VAR __DS0Core =
3         SUMMARIZECOLUMNS(
4             ROLLUPADDSUBTOTAL(ROLLUPGROUP('Product'[Category], 'Product'[Brand]), "IsGrandTotalRowTotal"),
5             "Margin", 'Sales'[Margin]
6         )
7
8     VAR __DS0PrimaryWindowed =
9         TOPN(502, __DS0Core, [IsGrandTotalRowTotal], 0, 'Product'[Category], 1, 'Product'[Brand], 1)
10
11 EVALUATE
12     __DS0PrimaryWindowed
13
14 ORDER BY
15     [IsGrandTotalRowTotal] DESC, 'Product'[Category], 'Product'[Brand]
```

Results | Result 1 of 1 |  Copy

	Product[Category]	Product[Brand]	[IsGrandTotalRowTotal]	[Margin]
1			True	1267049808.67
2	Audio	Contoso	False	5547636.55
3	Audio	Northwind Traders	False	5276681.88
4	Audio	Wide World Importers	False	21868862.18
5	Cameras and camcorders	A. Datum	False	28002980.67
6	Cameras and camcorders	Contoso	False	19656707.45
7	Cameras and camcorders	Fabrikam	False	71111794.44
8	Cell phones	Contoso	False	22490172.07
9	Cell phones	The Phone Company	False	174204193.03
10	Computers	Adventure Works	False	189547913.31

Query 1

Success (27.3 ms) Query 1 of 1 Result 1 of 1 4 columns, 30 rows

Query from Visual

Query View

- Queries are saved with the model
- Share or preserve slow query
- First exposure to DAX Queries
- Query View has many uses – i.e. validations
- Not good for diagnosing slow queries

DAX Query View for Web

- Just Announced

<https://powerbi.microsoft.com/en-us/blog/deep-dive-into-dax-query-view-for-web/>

- Use with models in the Workspaces

Power BI Demo Performance Analyzer

Performance Analyzer

- Informed Decisions
- Don't underestimate this tool
- You can isolate where the issue is
- Why spend all day optimizing DAX if bottleneck elsewhere



[This Photo](#) by Unknown Author is licensed under [CC BY](#)

DAX

What about that slow DAX Query?

Performance analyzer >>

Start recording Refresh visuals Stop

Clear Export

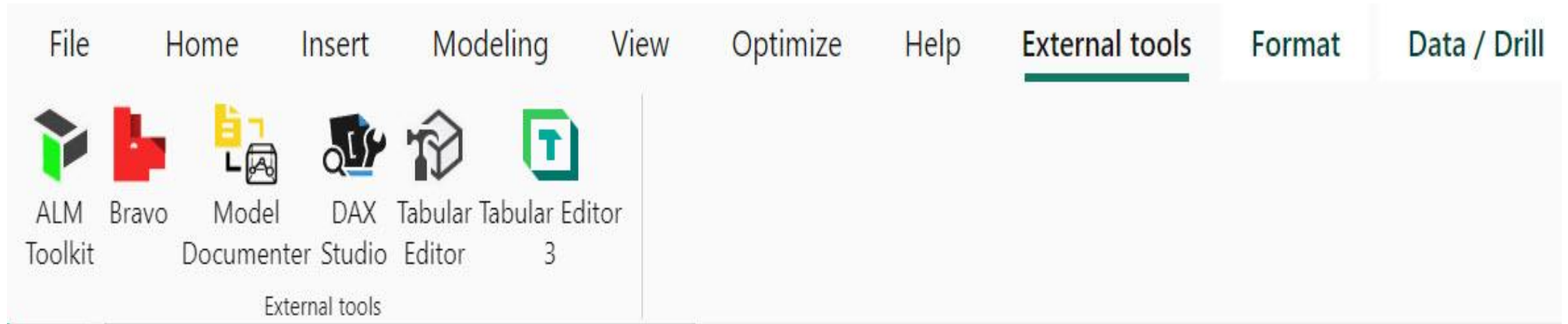
Name	Duration (ms) ↓
Refreshed visual	-
Table	61668
DAX query	61617
Visual display	23
Other	28
Copy query	
Run in DAX Query View	

A collection of various school and office supplies arranged on a solid light blue surface. The items include three wooden pencils (one sharpened), two yellow pencils, a black magnifying glass, a roll of white tape, a green paper clip, a blue sticky note, a grey T-shaped stapler, a small green eraser, a black ruler, a pair of glasses, a black and white marker, a teal highlighter, an orange stapler, four teal paper clips, a black and silver watch, a green water bottle, a black headset, a white pushpin, a yellow pushpin, a film strip, and a small portion of a ruler showing numbers 1 and 2.

Installation

- Full Install
- Power BI Desktop
- External Tools Tab

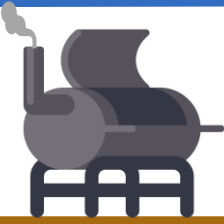
External Tools Tab



What if the DAX Query is Slow?



Our Journey



1. Intro

2. Performance Analyzer

3. DAX Studio

4. Tabular Editor

5. Conclusion

DAX Studio

- Author Measures and Queries
- Connect to Semantic Models
- Performance Tuning
 - Details on how the query is processed

Slow DAX Query

Performance analyzer >>

▶ Start recording ↺ Refresh visuals ⏹ Stop

⧫ Clear 📄 Export

Name	Duration (ms) ↓
↺ Refreshed visual	-
📄 Table	61668
DAX query	61617
Visual display	23
Other	28
📄 Copy query	
📄 DAX Run in DAX Query View	

Copy query

Performance analyzer >>

▶ Start recording ↺ Refresh visuals ⏹ Stop

⧫ Clear 📄 Export

Name	Duration (ms) ↓
↺ Refreshed visual	-
☐ Table	61668
DAX query	61617
Visual display	23
Other	28
📄 Copy query	
📄 Run in DAX Query View	

Copied

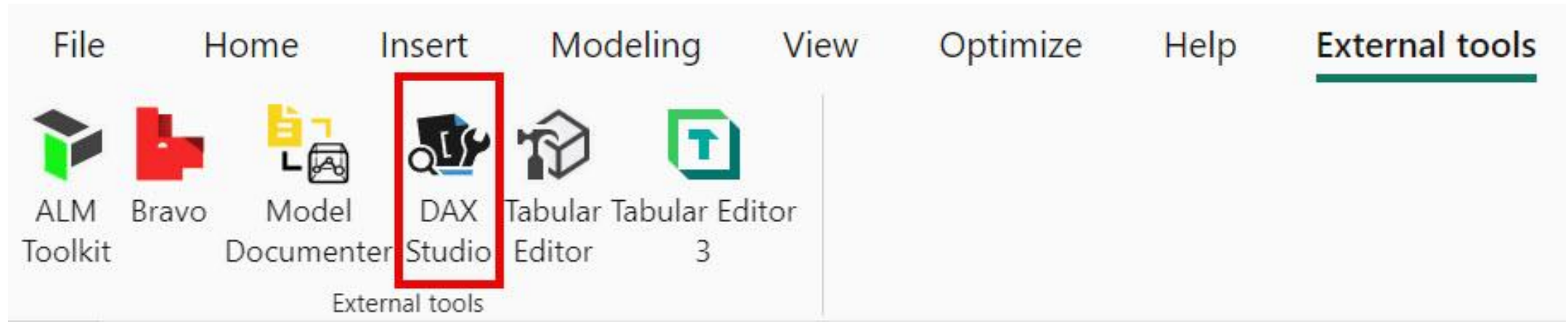
Performance analyzer >>

▶ Start recording ↺ Refresh visuals ⊞ Stop

◇ Clear 📄 Export

Name	Duration (ms) ↓
↺ Refreshed visual	-
☐ Table	61668
DAX query	61617
Visual display	23
Other	28
✓ Copied	
📄 Run in DAX Query View	

Power BI – Open DAX Studio



Paste in DAX Studio

DAX Studio - 3.0.11

File Home Advanced Help

Run Cancel Query Builder Clear Cache Clear on Run Results Output Edit Format Query To Upper To Lower Debug Commas Comment Uncomment Merge XML Find Replace Load Perf Data All Queries Query Plan Server Timings Connect Refresh Metadata

Metadata Functions DMV

Contoso1M_Docker_BPATest

Model

Search

- Customer
- Date
- Product
- Sales
- Store

```
1 DEFINE
2   VAR __DS0Core =
3     SUMMARIZECOLUMNS(
4       ROLLUPADDSUBTOTAL(ROLLUPGROUP('Product'[Category], 'Product'[Brand]), "IsGrandTotalRowTotal"),
5       "Margin", 'Sales'[Margin]
6     )
7
8   VAR __DS0Primarywindowed =
9     TOPN(502, __DS0Core, [IsGrandTotalRowTotal], 0, 'Product'[Category], 1, 'Product'[Brand], 1)
10
11 EVALUATE
12   __DS0Primarywindowed
13
14 ORDER BY
15   [IsGrandTotalRowTotal] DESC, 'Product'[Category], 'Product'[Brand]
```

Log Results History

	Start	Duration ⓘ	Message
ⓘ	16:34:01		Establishing Connection
ⓘ	16:34:01		Connected

Let's Run It

DAX Studio - 3.0.11

File **Home** Advanced Help

Run Cancel Query Builder Clear Cache Clear on Run Results Output Edit Format Query To Upper To Lower Debug Commas Comment Uncomment Merge XML Find Replace Load Perf Data All Queries Query Plan Server Timings Connect Refresh Metadata

Metadata Functions DMV

Contoso1M_Docker_BPATest

Model

Search

- Customer
- Date
- Product
- Sales
- Store

```
1 DEFINE
2   VAR __DS0Core =
3     SUMMARIZECOLUMNS(
4       ROLLUPADDSUBTOTAL(ROLLUPGROUP('Product'[Category], 'Product'[Brand]), "IsGrandTotalRowTotal"),
5       "Margin", 'Sales'[Margin]
6     )
7
8   VAR __DS0Primarywindowed =
9     TOPN(502, __DS0Core, [IsGrandTotalRowTotal], 0, 'Product'[Category], 1, 'Product'[Brand], 1)
10
11 EVALUATE
12   __DS0Primarywindowed
13
14 ORDER BY
15   [IsGrandTotalRowTotal] DESC, 'Product'[Category], 'Product'[Brand]
```

Log Results History

	Start	Duration ⓘ	Message
ⓘ	16:34:01		Establishing Connection
ⓘ	16:34:01		Connected

Results – Like Query View

FileHomeAdvancedHelp

Run

Cancel

Query Builder

Clear Cache

Clear on Run

Results

Cut

Copy

Paste

Format Query

To Upper

To Lower

Debug Commas

Comment

Uncomment

Merge XML

Find

Replace

Load Perf Data

All Queries

Query Plan

Server Timings

Connect

Refresh Metadata

QueryCacheOutputEditFormatFindPower BITracesConnection

MetadataFunctionsDMV

Contoso1M_Docker_BPATest

Model

Search

Customer

Date

Product

Sales

Store

```
1 DEFINE
2   VAR __DS0Core =
3     SUMMARIZECOLUMNS(
4       ROLLUPADDISSUBTOTAL(ROLLUPGROUP('Product'[Category], 'Product'[Brand]), "IsGrandTotalRowTotal"),
5       "Margin", 'Sales'[Margin]
6     )
7
8   VAR __DS0Primarywindowed =
9     TOPN(502, __DS0Core, [IsGrandTotalRowTotal], 0, 'Product'[Category], 1, 'Product'[Brand], 1)
10
11 EVALUATE
12   __DS0Primarywindowed
13
14 ORDER BY
15   [IsGrandTotalRowTotal] DESC, 'Product'[Category], 'Product'[Brand]
```

LogResultsHistoryServer Timings

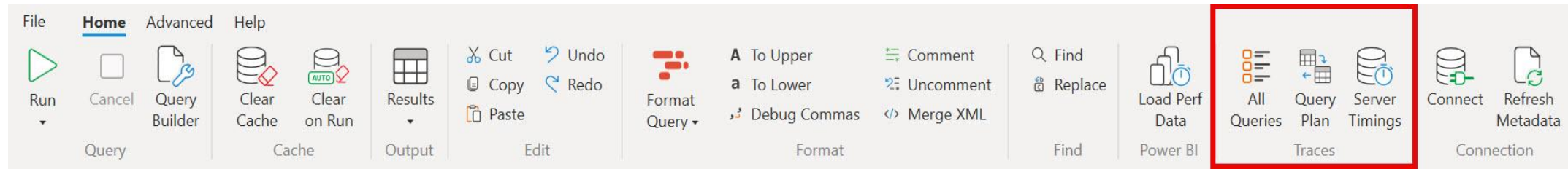
Category	Brand	IsGrandTotalRowTotal	Margin
		True	1267049808.6663
Audio	Contoso	False	5547636.5545
Audio	Northwind Traders	False	5276681.882
Audio	Wide World Importers	False	21868862.1847
Cameras and camcorders	A. Datum	False	28002980.6738
Cameras and camcorders	Contoso	False	19656707.4536
Cameras and camcorders	Fabrikam	False	71111794.4404
Cell phones	Contoso	False	22490172.075
Cell phones	The Phone Company	False	174204193.028
Computers	Adventure Works	False	189547913.3088

Query1.dax* X

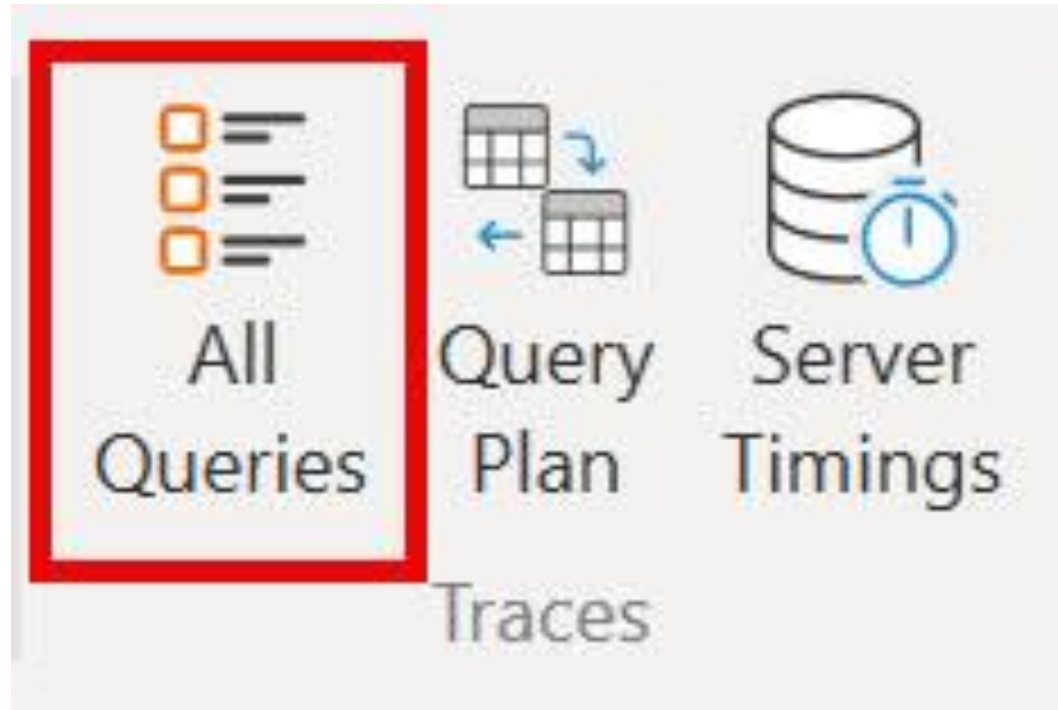
Ready

Ln 15, Col 50localhost:5319316.0.126.2710230 rows00:00.0

Traces



DAX Studio – All Queries



Like SQL Server Profiler

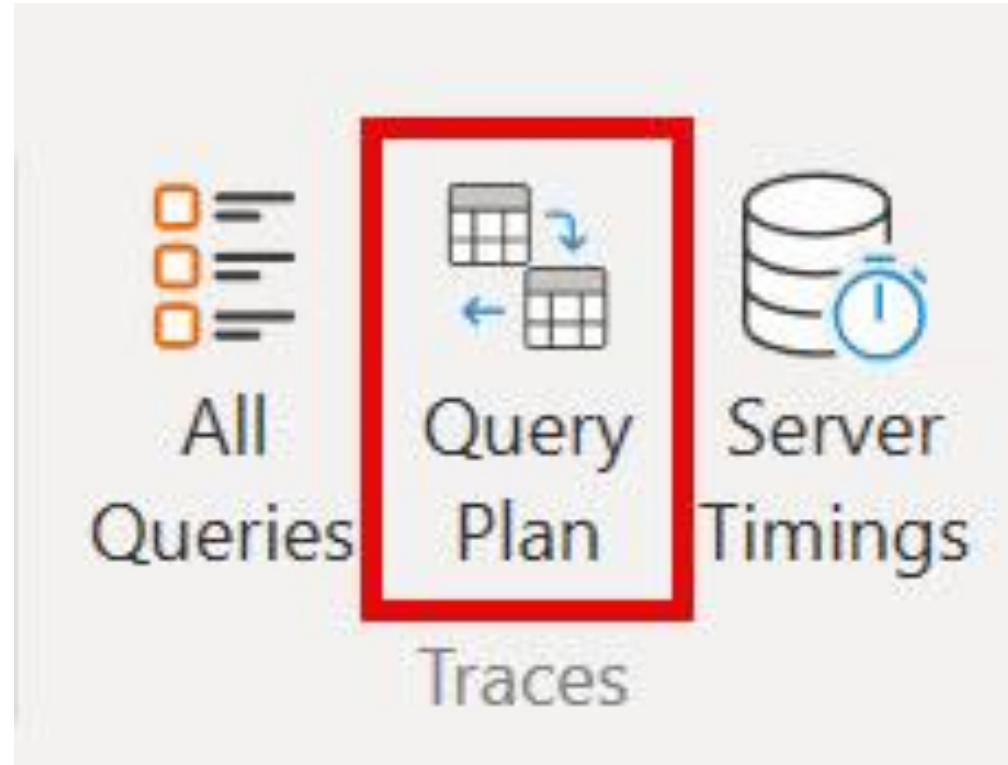
DAX Studio – All Queries

Log Results History **All Queries**

Record Pause Stop Clear Copy Export Info Filters

StartTime	Type	Duration	User	Database	Query
10:12:54	DAX	10ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = SUMMARIZECOLUMNS(ROLLUPADDISSUBTOTAL(ROLLUPGROUP('Product'[Category], 'Product'[Brand]), "IsGrandTotalRowTotal"), "Margin", 'Sa...
10:12:51	DAX	0ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = SUMMARIZE('Product', 'Product'[Category], 'Product'[Brand]) VAR __DS0PrimaryWindowed = TOPN(501, __DS0Core, 'Product'[Category], 1, 'Pro...
10:12:50	DAX	2ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = DISTINCT('Product'[Category]) VAR __DS0PrimaryWindowed = TOPN(501, __DS0Core, 'Product'[Category], 1) EVALUATE __DS0PrimaryWindowe...
10:11:45	DAX	187ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = SUMMARIZECOLUMNS(ROLLUPADDISSUBTOTAL(ROLLUPGROUP('Date'[Year Month], 'Product'[Brand]), "IsGrandTotalRowTotal"), "Total_BIG_Or...
10:11:39	DAX	9ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = SELECTCOLUMNS(KEEPFILTERS(FILTER(KEEPFILTERS(SUMMARIZECOLUMNS('Date'[Year Month], 'Product'[Brand], "CountRowsSales", COUNT...
10:11:24	DAX	6ms	MSI\roman	Contoso1M_Docker_BPATest	EVALUATE ROW("MinBrand", CALCULATE(MIN('Product'[Brand])))
10:10:33	DAX	31,984ms	MSI\roman	Contoso1M_Docker_BPATest	DEFINE VAR __DS0Core = SUMMARIZECOLUMNS(ROLLUPADDISSUBTOTAL(ROLLUPGROUP('Date'[Year Month], 'Product'[Brand]), "IsGrandTotalRowTotal"), "Total_BIG_Or...

Query Plan



DAX Studio

- Query Plans
 - Logical
 - Physical

DAX Studio – Query Plan

LogResultsHistoryServer TimingsQuery Plan

RecordPauseStopClearExportInfo

Line	Records	Physical Query Plan
1		PartitionIntoGroups: IterPhyOp LogOp=Order IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[]") #Groups=1 #Rows=120
2	1	AggregationSpool<Order>: SpoolPhyOp #Records=1
3		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[]")
4		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[]")
5		Union: IterPhyOp LogOp=Union IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[]")
6		GroupSemiJoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])
7	119	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq IterCols(0)('Date'[Year Month]) #Records=119 #KeyCols=70 #ValueCols=1
8	119	ProjectionSpool<ProjectFusion<Copy>>: SpoolPhyOp #Records=119
9		Cache: IterPhyOp #FieldCols=1 #ValueCols=1
10		GroupSemiJoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])
11	1	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq #Records=1 #KeyCols=70 #ValueCols=1

Line	Logical Query Plan
1	_DS0Core: Union: RelLogOp VarName=__DS0Core DependOnCols() 0-3 RequiredCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[]")
2	GroupSemiJoin: RelLogOp DependOnCols() 0-2 RequiredCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])
3	Scan_Vertipaq: RelLogOp DependOnCols() 0-0 RequiredCols(0)('Date'[Year Month])
4	Constant: ScaLogOp DependOnCols() Boolean DominantValue=false
5	Calculate: ScaLogOp MeasureRef=[Total BIG Orders Good] DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK
6	DistinctCount_Vertipaq: ScaLogOp DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK
7	Scan_Vertipaq: RelLogOp DependOnCols(0)('Date'[Year Month]) 2-2 RequiredCols(0)('Date'[Year Month])
8	Filter_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])
9	Scan_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])
10	GreaterThan: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Boolean DominantValue=NONE
11	'Sales'[Net Price]: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Currency DominantValue=NONE

What the French Toast

Query Plan – One Useful Thing

LogResultsHistoryServer TimingsQuery Plan

RecordPauseStopClearExportInfo

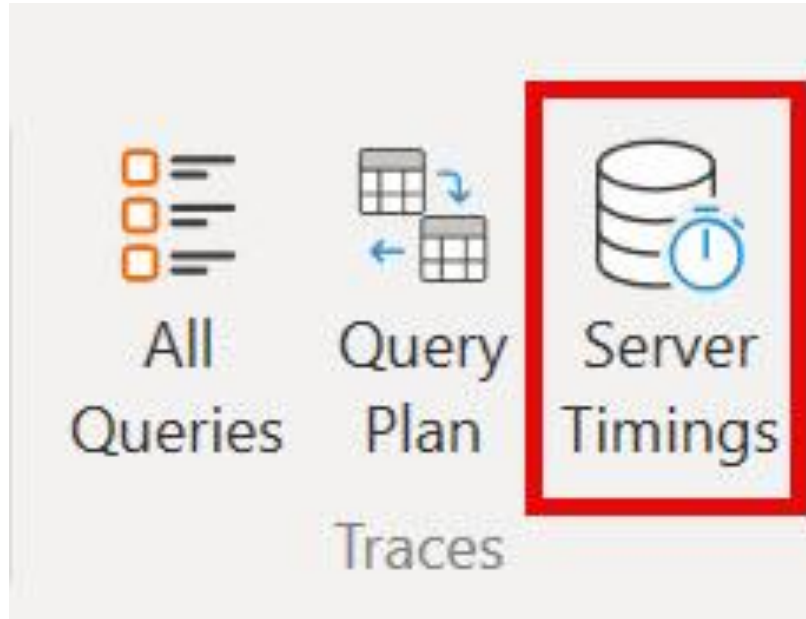
Line	Records	Physical Query Plan
1		PartitionIntoGroups: IterPhyOp LogOp=Order IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good], "[[]] #Groups=1 #Rows=120
2	1	AggregationSpool<Order>: SpoolPhyOp #Records=1
3		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good], "[[]]
4		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good], "[[]]
5		Union: IterPhyOp LogOp=Union IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good], "[[]]
6		GroupSemiJoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good])
7	119	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq IterCols(0)('Date'[Year Month]) #Records=119 #KeyCols=70 #ValueCols=1
8	119	ProjectionSpool<ProjectFusion<Copy>>: SpoolPhyOp #Records=119
9		Cache: IterPhyOp #FieldCols=1 #ValueCols=1
10		GroupSemiJoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good])
11	1	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq #Records=1 #KeyCols=70 #ValueCols=1

Line	Logical Query Plan
1	_DS0Core: Union: RelLogOp VarName=__DS0Core DependOnCols() 0-3 RequiredCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good], "[[]]
2	GroupSemiJoin: RelLogOp DependOnCols() 0-2 RequiredCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal], "[Total_BIG_Orders_Good])
3	Scan_Vertipaq: RelLogOp DependOnCols() 0-0 RequiredCols(0)('Date'[Year Month])
4	Constant: ScaLogOp DependOnCols() Boolean DominantValue=false
5	Calculate: ScaLogOp MeasureRef=[Total BIG Orders Good] DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK
6	DistinctCount_Vertipaq: ScaLogOp DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK
7	Scan_Vertipaq: RelLogOp DependOnCols(0)('Date'[Year Month]) 2-2 RequiredCols(0)('Date'[Year Month])
8	Filter_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])
9	Scan_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])
10	GreaterThan: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Boolean DominantValue=NONE
11	'Sales'[Net Price]: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Currency DominantValue=NONE

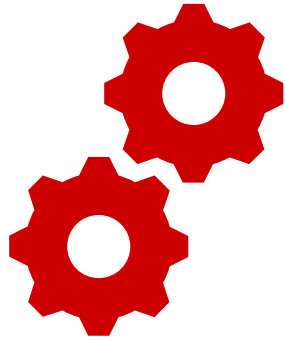
DAX Studio – Records

Log	Results	History	Server Timings	Query Plan	
Record	Pause	Stop	Clear	Export	Info
Line	Records	Physical Query Plan			
1		PartitionIntoGroups: IterPhyOp LogOp=Order IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[[]]") #Groups=1 #Rows=120			
2	1	AggregationSpool<Order>: SpoolPhyOp #Records=1			
3		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[[]]")			
4		Proxy: IterPhyOp LogOp=TableVarProxy IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[[]]")			
5		Union: IterPhyOp LogOp=Union IterCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[[]]")			
6		GroupSemijoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])			
7	119	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq IterCols(0)('Date'[Year Month]) #Records=119 #KeyCols=70 #ValueCols=1			
8	119	ProjectionSpool<ProjectFusion<Copy>>: SpoolPhyOp #Records=119			
9		Cache: IterPhyOp #FieldCols=1 #ValueCols=1			
10		GroupSemijoin: IterPhyOp LogOp=GroupSemiJoin IterCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])			
11	1	Spool_Iterator<SpoolIterator>: IterPhyOp LogOp=DistinctCount_Vertipaq IterCols(0)('Date'[Year Month]) #Records=1 #KeyCols=70 #ValueCols=1			
Line		Logical Query Plan			
1		_DS0Core: Union: RelLogOp VarName=__DS0Core DependOnCols() 0-3 RequiredCols(0, 1, 2, 3)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good]", "[[]]")			
2		GroupSemiJoin: RelLogOp DependOnCols() 0-2 RequiredCols(0, 1, 2)('Date'[Year Month], "[IsGrandTotalRowTotal]", "[Total_BIG_Orders_Good])			
3		Scan_Vertipaq: RelLogOp DependOnCols() 0-0 RequiredCols(0)('Date'[Year Month])			
4		Constant: ScaLogOp DependOnCols() Boolean DominantValue=false			
5		Calculate: ScaLogOp MeasureRef=[Total BIG Orders Good] DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK			
6		DistinctCount_Vertipaq: ScaLogOp DependOnCols(0)('Date'[Year Month]) Integer DominantValue=BLANK			
7		Scan_Vertipaq: RelLogOp DependOnCols(0)('Date'[Year Month]) 2-2 RequiredCols(0)('Date'[Year Month])			
8		Filter_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])			
9		Scan_Vertipaq: RelLogOp DependOnCols() 1-1 RequiredCols(1)('Sales'[Net Price])			
10		GreaterThan: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Boolean DominantValue=NONE			
11		'Sales'[Net Price]: ScaLogOp DependOnCols(1)('Sales'[Net Price]) Currency DominantValue=NONE			

DAX Studio – Server Timings



A Tale of Two Engines



Formula Engine

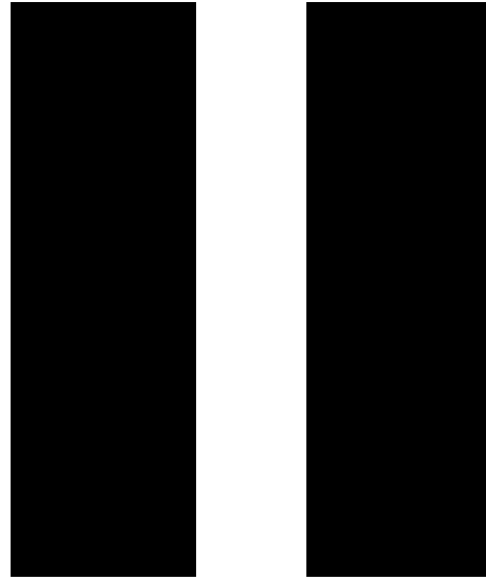


Storage Engine

Formula Engine

- Conductor
 - Does not cache
 - Single threaded
 - Complex Operations
- 

Pause



Storage Engine

- Not referring to:
 - Import Mode
 - Loading data when processing the model
 - Query Folding when pulling data from source
- Data has already been loaded to Vertipaq
- Accessing that data for Queries and Visuals

Storage Engine

- This is where we want most of the time spent
- Ability to cache
- Ability to Multithread
- The type of Operations available depends on the storage engine
 - Vertipaq is limited

Beyond the Scope



Storage Engine(s)

- Different Storage Engines
 - Vertipaq (Import)
 - Direct Query SQL (SQL Server, Snowflake)
- Combination of these
 - Import with Direct Query SQL
 - Direct Query over Tabular

More than Vertipaq (Import)

- Going beyond Import increases the complexity
 - For Example:
 - Date Table is Import and the rest of the model is Direct Query
- Need to do performance testing

Storage Engine Capability

- Vertipaq
 - Needs Formula Engine for IF Statement
- SQL Server
 - Can push IF equivalent to SQL Server

Vertipaq

- Column Based
- Encoded
- Compressed

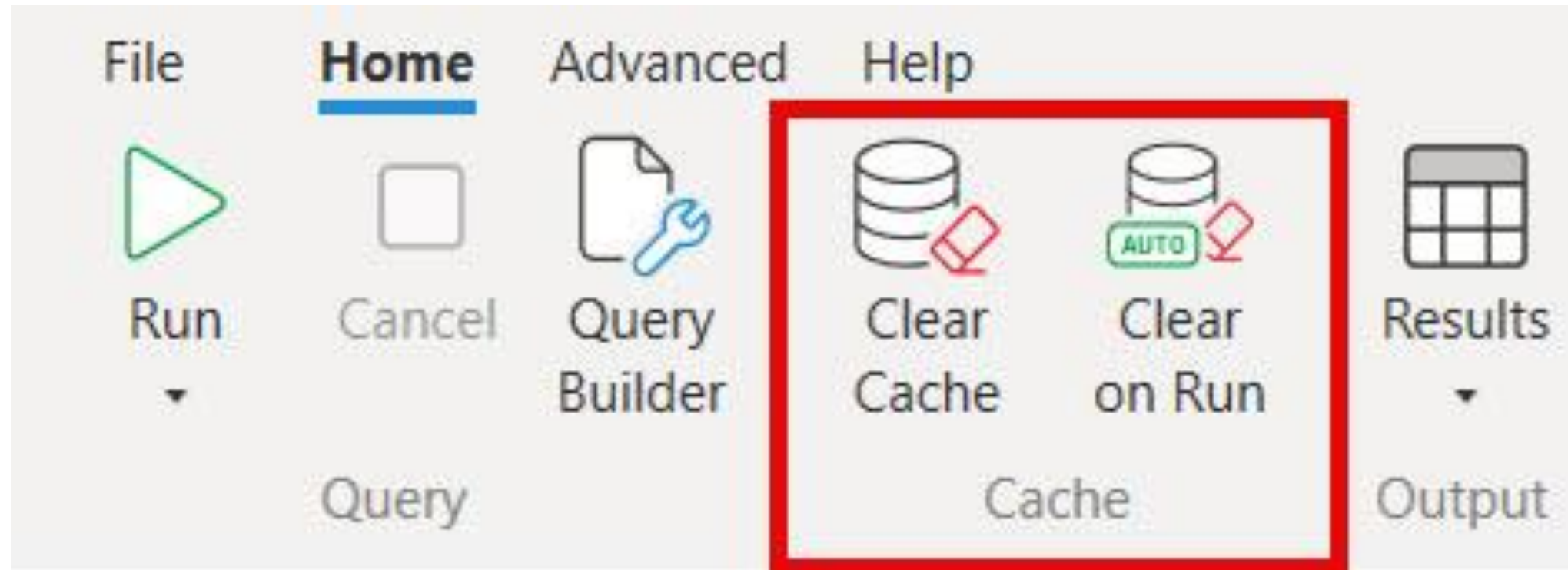
Vertipaq

- Restaurant
- Local Storeroom that is organized

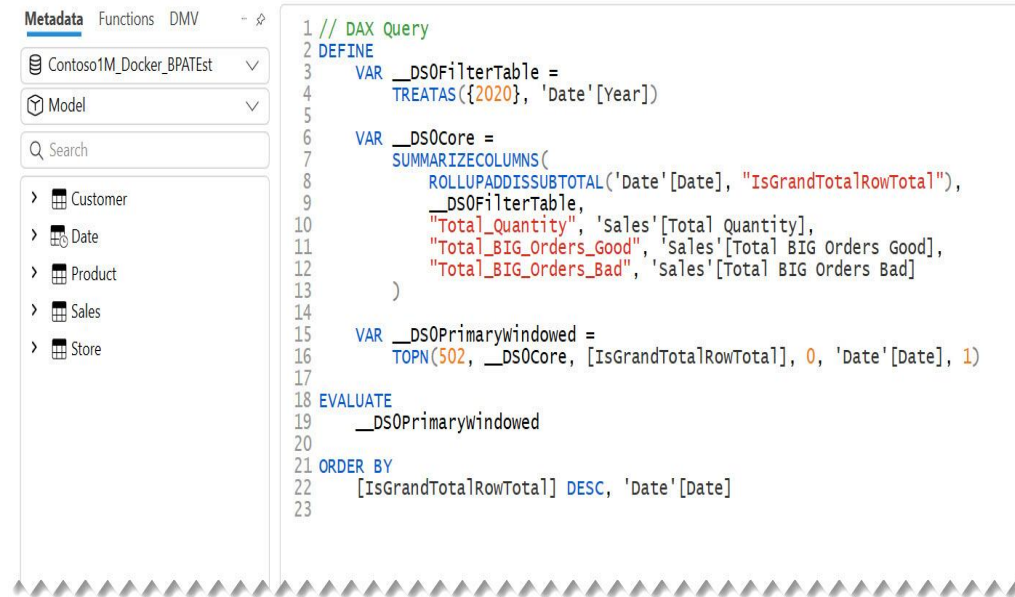
Direct Query

- Restaurant
 - Supplier brings what I request as needed
 - Offers pre-made items but at a cost

Cache

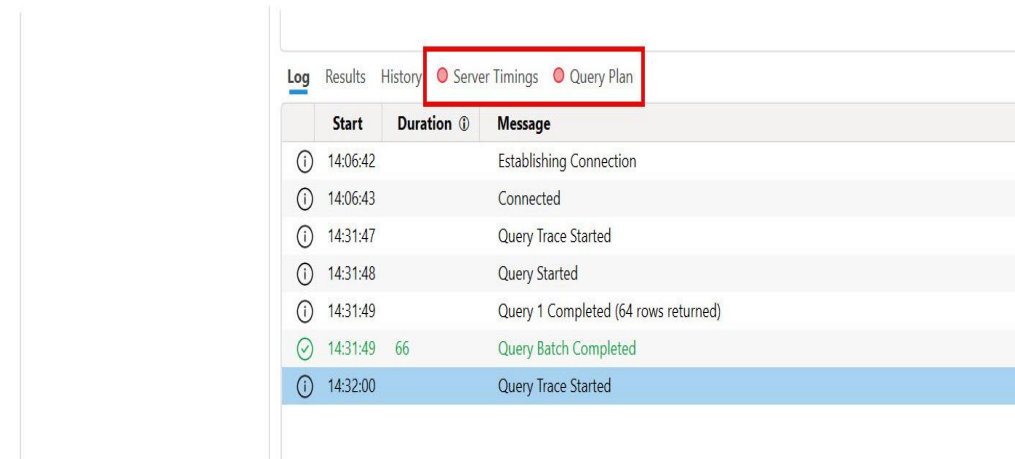


Location of Query and Server Timings



The screenshot shows the SQL Server Data Tools (SSDT) interface. On the left, the 'Metadata' pane is active, displaying a tree view of the 'Contoso1M_Docker_BPATest' model. The tree includes folders for 'Customer', 'Date', 'Product', 'Sales', and 'Store'. The main query editor on the right contains a DAX query script:

```
1 // DAX Query
2 DEFINE
3     VAR __DS0FilterTable =
4         TREATAS({2020}, 'Date'[Year])
5
6     VAR __DS0Core =
7         SUMMARIZECOLUMNS(
8             ROLLUPADISSUBTOTAL('Date'[Date], "IsGrandTotalRowTotal"),
9             __DS0FilterTable,
10            "Total_Quantity", 'Sales'[Total Quantity],
11            "Total_BIG_Orders_Good", 'Sales'[Total BIG Orders Good],
12            "Total_BIG_Orders_Bad", 'Sales'[Total BIG Orders Bad]
13        )
14
15     VAR __DS0Primarywindowed =
16         TOPN(502, __DS0Core, [IsGrandTotalRowTotal], 0, 'Date'[Date], 1)
17
18 EVALUATE
19     __DS0Primarywindowed
20
21 ORDER BY
22     [IsGrandTotalRowTotal] DESC, 'Date'[Date]
23
```



The screenshot shows the 'Log' window in SSDT, which displays a list of events. The 'Log' tab is selected, and the 'Server Timings' and 'Query Plan' filters are checked. The log entries are as follows:

	Start	Duration	Message
(i)	14:06:42		Establishing Connection
(i)	14:06:43		Connected
(i)	14:31:47		Query Trace Started
(i)	14:31:48		Query Started
(i)	14:31:49		Query 1 Completed (64 rows returned)
(✓)	14:31:49	66	Query Batch Completed
(i)	14:32:00		Query Trace Started

DAX – Return Product Table

EVALUATE

'Product'

Storage Engine Query

1 EVALUATE
2 'Product'

Log Results History **Server Timings**

Record Pause Stop Clear Copy Export Info Copy SE query Scan Cache Batch Internal

Total	SE CPU	Line	Subclass	Duration	CPU	Par.	Rows	KB	Timeline	Query
37 ms	0 ms x0.0	2	Scan	1	0		2,517	148		SELECT 'Product'[RowNumber], 'Product'[ProductKey],

FE 36 ms 97.3% SE 1 ms 2.7%

SE Queries 1 SE Cache 0 0.0%

```
SET DC_KIND="AUTO";  
SELECT  
    'Product'[RowNumber],  
    'Product'[ProductKey],  
    'Product'[Product Code],  
    'Product'[Product Name],  
    'Product'[Manufacturer],  
    'Product'[Brand],  
    'Product'[Color],  
    'Product'[Weight Unit Measure],  
    'Product'[Weight],  
    'Product'[Unit Cost],  
    'Product'[Unit Price],  
    'Product'[Subcategory Code],  
    'Product'[Subcategory],  
    'Product'[Category Code],  
    'Product'[Category]  
FROM 'Product';
```

Estimated size: rows = 2,517 bytes = 151,020

Vertipaq – xmSQL

SELECT

```
'Product'[RowNumber],  
'Product'[ProductKey],  
'Product'[Product Code],  
'Product'[Product Name],  
'Product'[Manufacturer],  
'Product'[Brand],  
'Product'[Color],  
'Product'[Weight Unit Measure],  
'Product'[Weight],  
'Product'[Unit Cost],  
'Product'[Unit Price],  
'Product'[Subcategory Code],  
'Product'[Subcategory],  
'Product'[Category Code],  
'Product'[Category]
```

FROM 'Product';

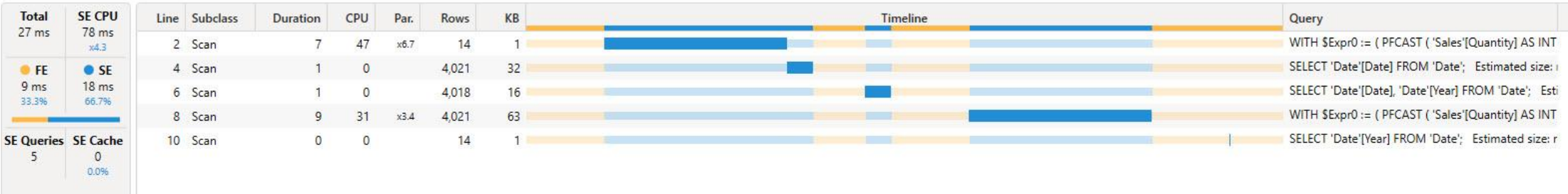
Server Timings - Timeline

This one is mainly in Formula Engine
Blue shows where Storage Engine is in the timeline

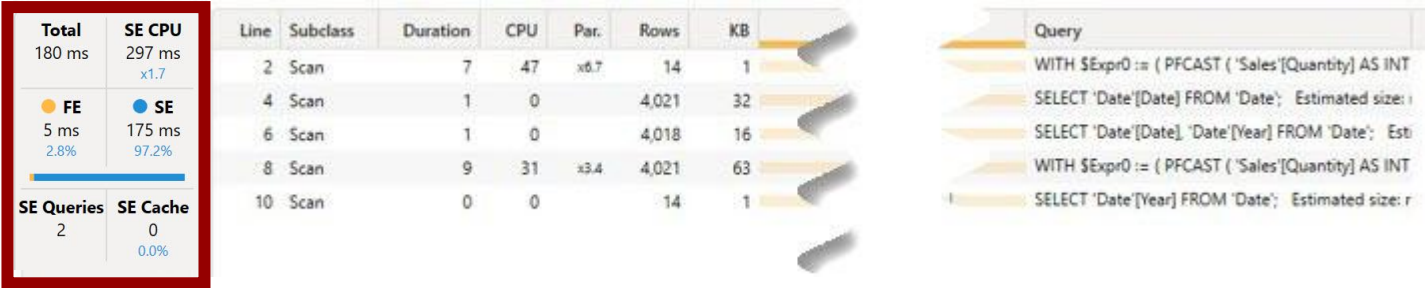
Line	Subclass	Duration	CPU	Par.	Rows	KB	Timeline	Query
2	Scan	0	0		4,021	32		SELECT 'Date'[Date] FROM 'Date'; Estimated size: rows = 4,021 bytes = 32,168
4	Scan	0	0		4,021	32		SELECT 'Date'[Date] FROM 'Date'; Estimated size: rows = 4,021 bytes = 32,168
6	Scan	0	0		4,021	63		SELECT 'Date'[Date], MAX ('Date'[Date]) FROM 'Date'; Estimated size: rows = 4,021 bytes = 64,336
8	Scan	7	31	x4.4	4,021	63		WITH \$Expr0 := (PFCAST ('Sales'[Quantity] AS INT) * PFCAST ('Sales'[Net Price] AS INT)) SELECT 'Date'[Date], SUM (@\$Expr0) FR

Server Timings - Timeline

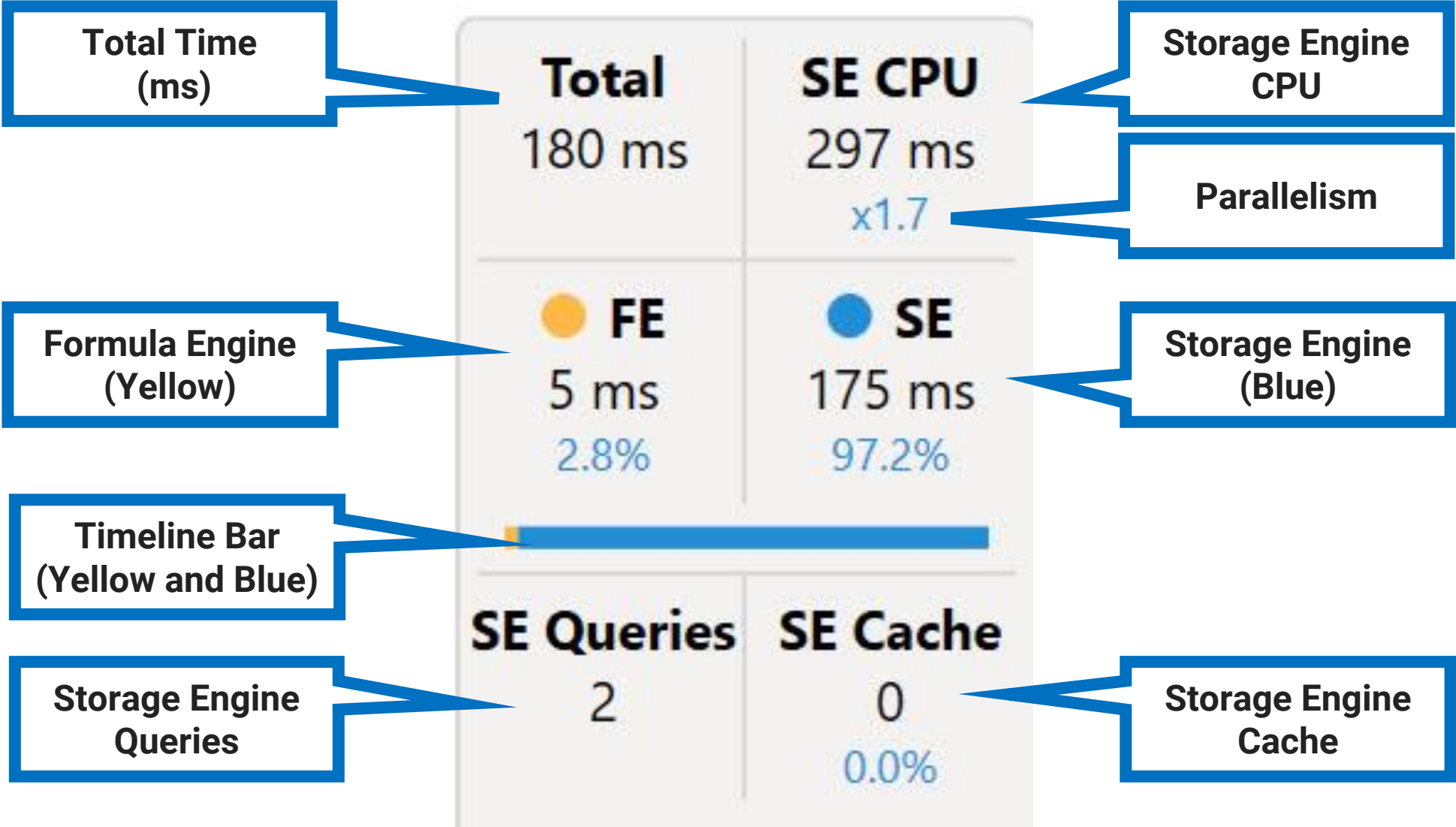
This one 33.3% in Formula Engine
Blue shows where Storage Engine is in the timeline.



Server Timings



Server Timings



Simple way to increase Query Time

- Filter by the full table

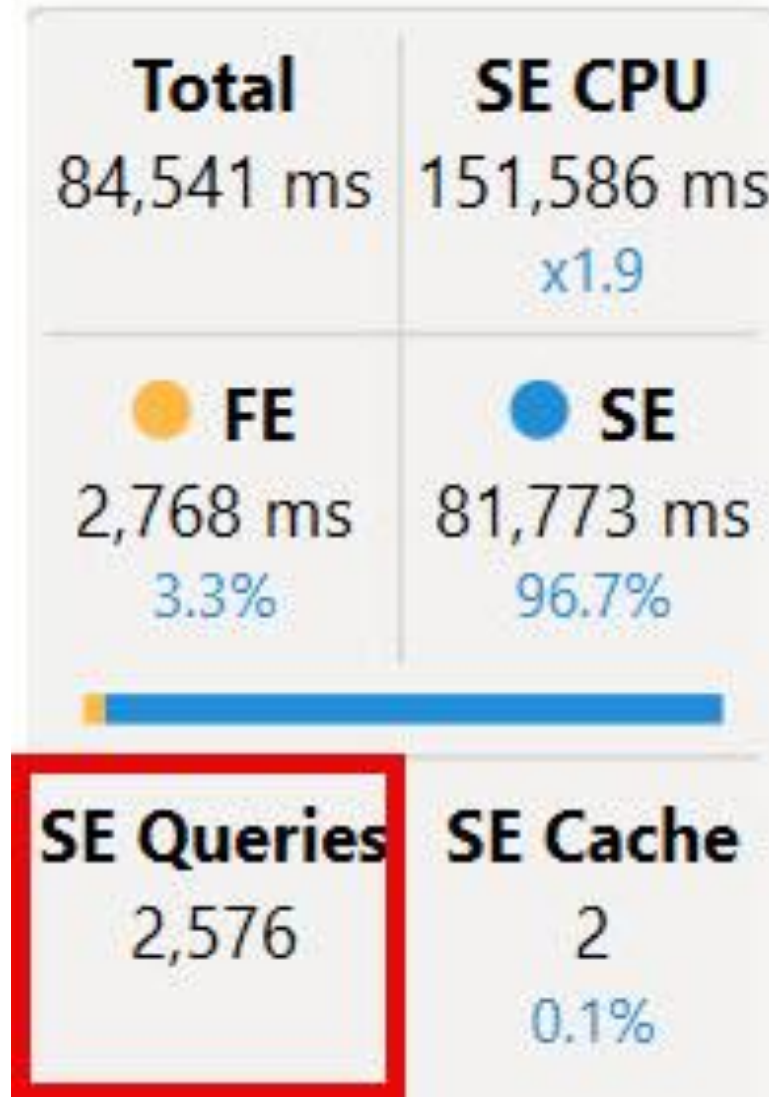
CALCULATE (

[Sales Amount],

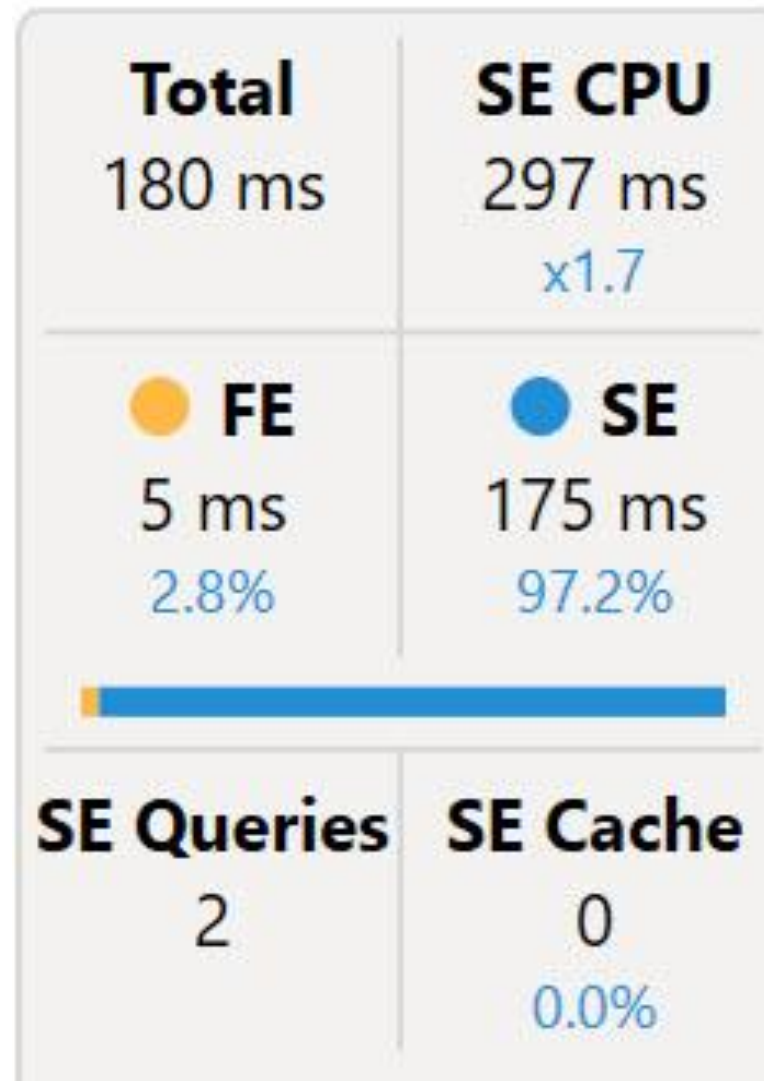
FILTER (Sales, Sales[Quantity] > 1

)

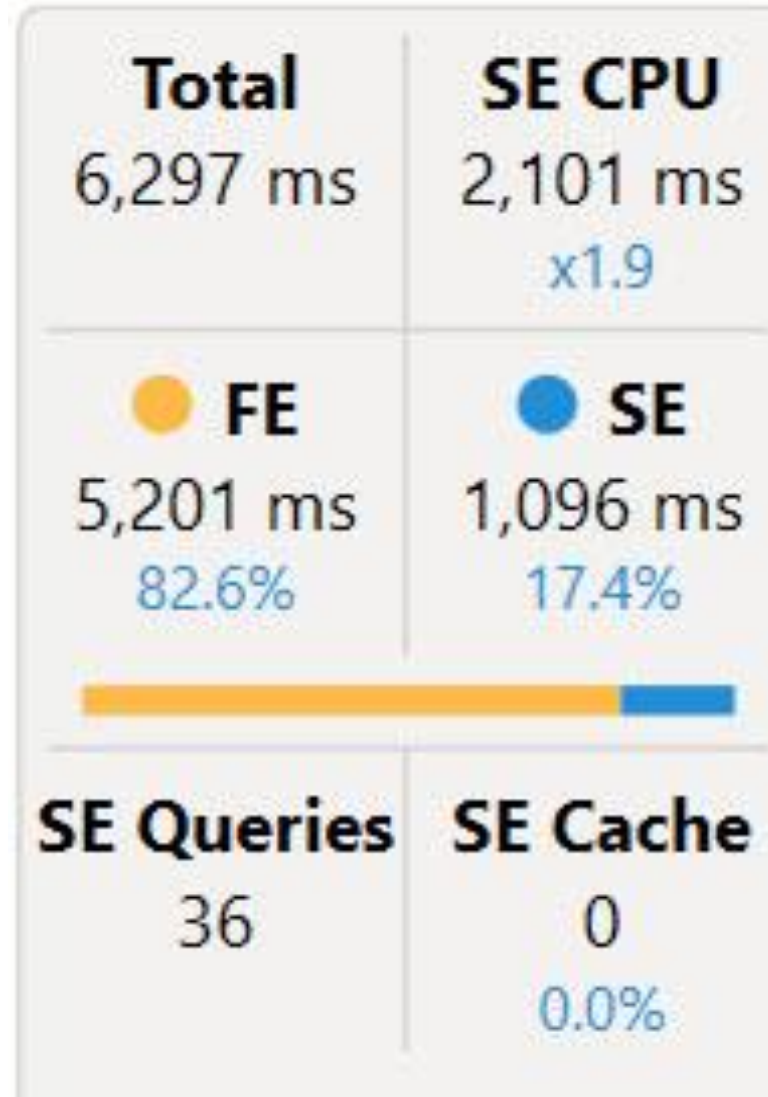
DAX Studio – Server Timings – Bad



DAX Studio – Server Timings – Good



DAX Studio – Heavy Formula Engine



Ways to invoke Formula Engine (Vertipaq)

- IF Statements
- CONCATENATE

CallbackDataID

WHERE

```
( COALESCE ( [CallbackDataID] ( IF (
'Product'[Color] = "Blue", 1,0 ) ) ]
( PFDATAID ( 'Product'[Color] ) ) ) <> 0 );
```

Direct Query Model

- Direct Query
 - How many queries are being sent to Source

Power BI Demo

DAX Studio

Proactive Measures*

(*no pun intended)

- How do we find issues with the Model and DAX ahead of time

Our Journey



1. Intro
2. Performance Analyzer
3. DAX Studio
- 4. Tabular Editor**
5. Conclusion

Tabular Editor

- Main Uses
 - Develop the model
 - Make changes
 - Audit the model
 - Best Practice Analyzer

Tabular Editor 2.x

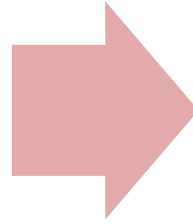
- Free version
- Version listed in DP-600 Study Guide

Tabular Editor 3.x

- Paid Version
- Extra Features
 - DAX Debugger
 - Script DAX

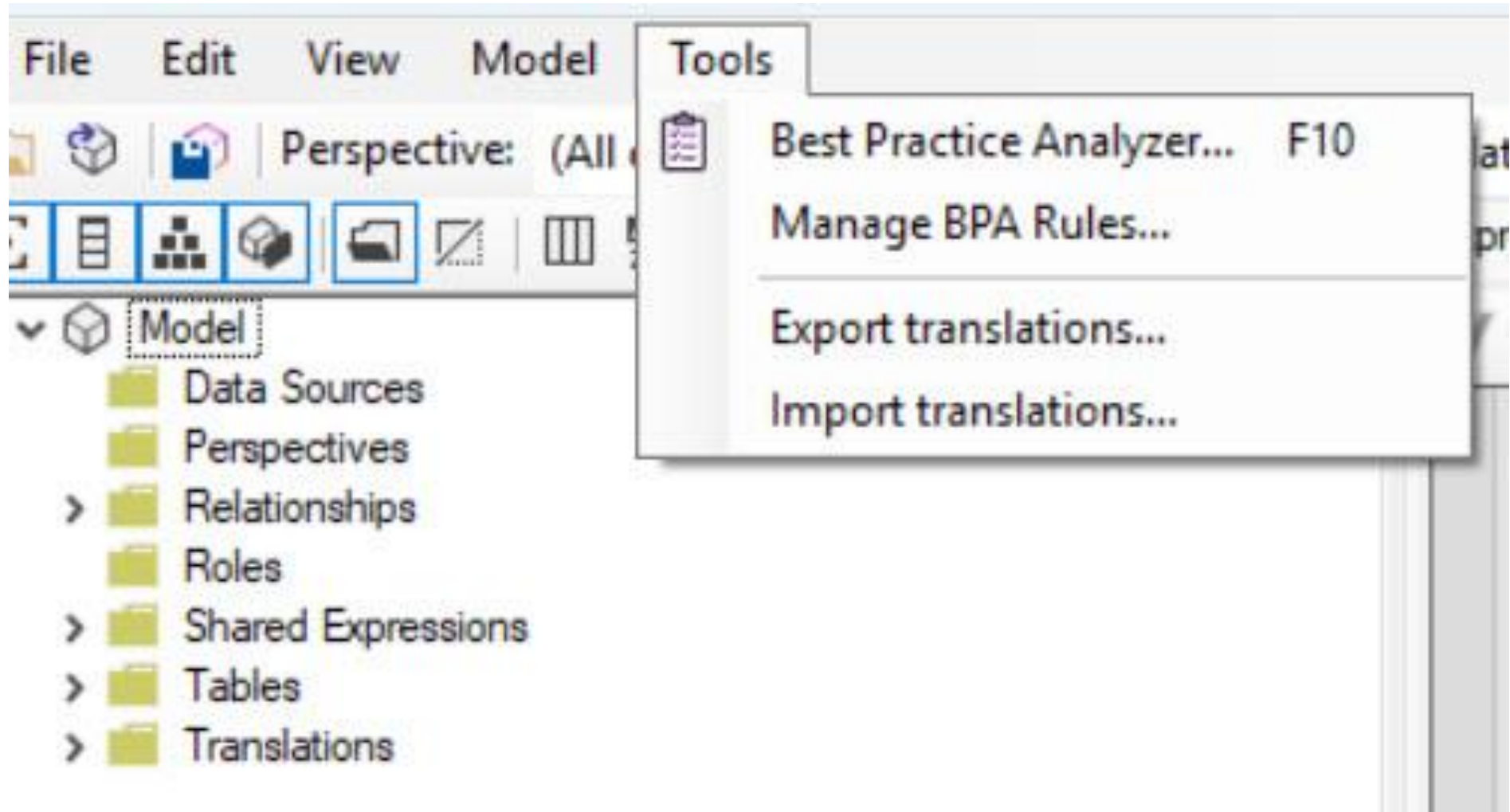
Tabular Editor 2

Optimize a semantic
model by using
Tabular Editor 2

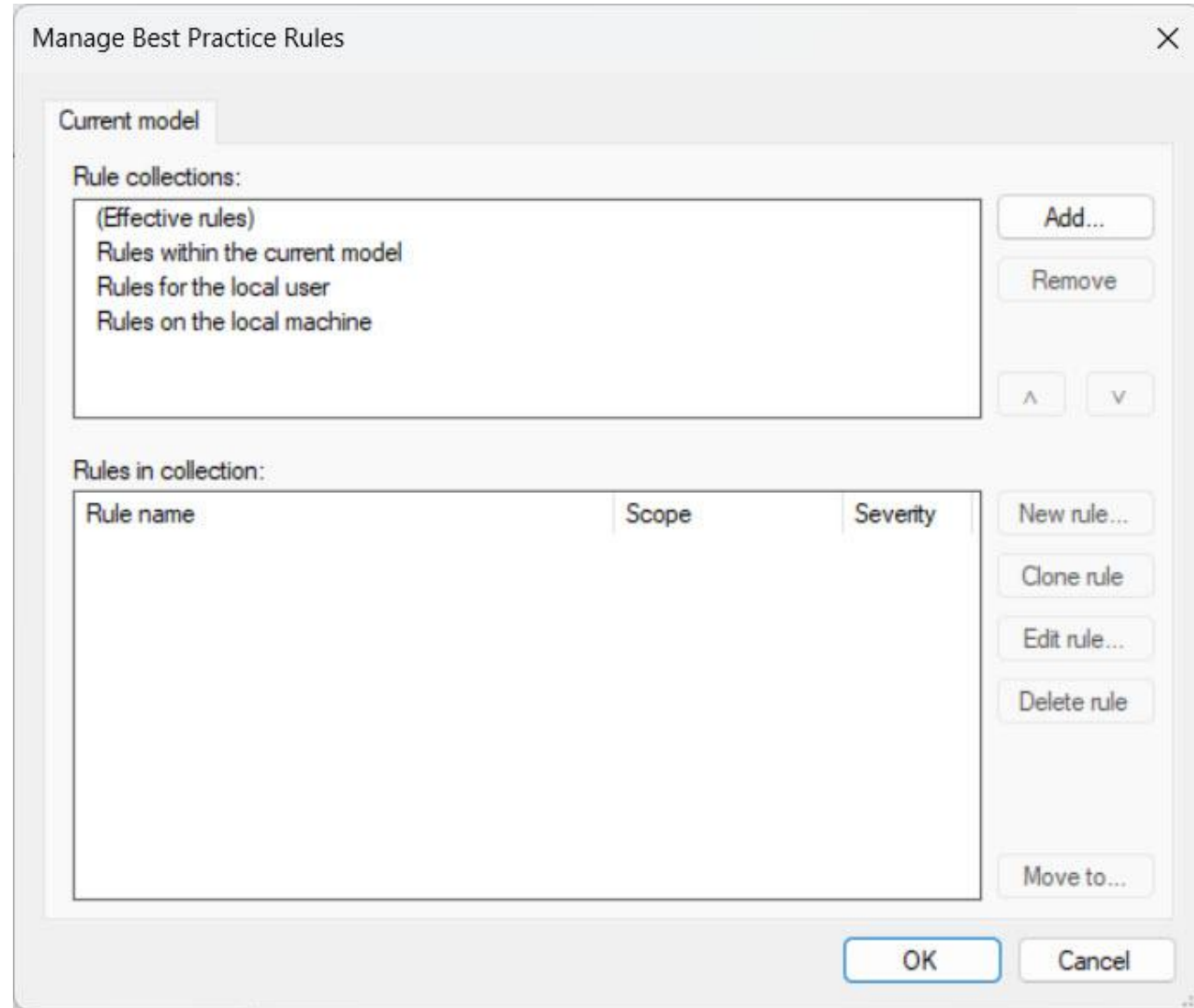


Best Practice Analyzer

Best Practice Analyzer Location



Add and Manage Rules



Best Practice Analyzer – C# Script Download

```
System.Net.WebClient w = new System.Net.WebClient();

string path = System.Environment.GetFolderPath(System.Environment.SpecialFolder.LocalApplicationData);
string url = "https://raw.githubusercontent.com/microsoft/Analysis-Services/master/BestPracticeRules/BPARules.json";
string version = System.Windows.Forms.Application.ProductVersion.Substring(0,1);
string downloadLoc = path + @"\" + TabularEditor + "BPARules.json";

if (version == "3")
{
    downloadLoc = path + @"\" + TabularEditor3 + "BPARules.json";
}

w.DownloadFile(url, downloadLoc);

/*
// Italian
string url = "https://raw.githubusercontent.com/microsoft/Analysis-Services/master/BestPracticeRules/Italian/BPARules.json";

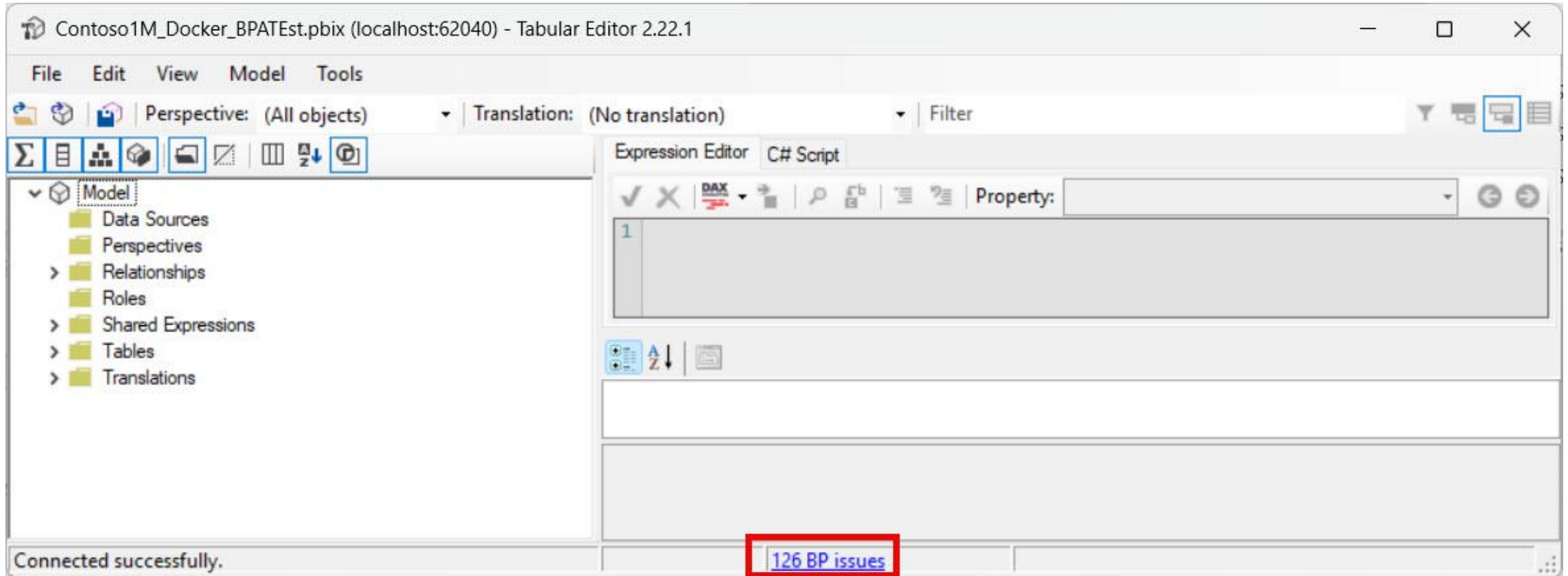
// Japanese
string url = "https://raw.githubusercontent.com/microsoft/Analysis-Services/master/BestPracticeRules/Japanese/BPARules.json";

// Spanish
string url = "https://raw.githubusercontent.com/microsoft/Analysis-Services/master/BestPracticeRules/Spanish/BPARules.json";
*/
```

Nope – Not Running a Script

- Alternate Method
 - Microsoft Analysis-Services – Best Practice Rules
 - <https://github.com/microsoft/Analysis-Services/tree/master/BestPracticeRules>
 - Other languages besides English

BPA – Rule Exceptions for model



BPA – Rule Exceptions for model

Object	Type	Severity
+ [Performance] Do not use floating point data types (2 objects)		2
+ [Performance] Set IsAvailableInMdx to false on non-attribute columns (13 objects)		2
- [Performance] Remove redundant columns in related tables (2 objects)		2
'Sales'[Unit Price]	Column	
'Sales'[Unit Cost]	Column	
- [DAX Expressions] Filter column values with proper syntax (1 object)		2
[Total BIG Orders Bad]	Measure	
+ [Maintenance] Remove unnecessary columns (5 objects)		2
+ [Maintenance] Visible objects with no description (66 objects)		1
+ [Naming Conventions] Partition name should match table name for single partition tables (5 objects)		1
+ [Formatting] Provide format string for "Date" columns (5 objects)		1
+ [Formatting] Provide format string for measures (1 object)		3
+ [Formatting] Percentages should be formatted with thousands separators and 1 decimal (1 object)		2
+ [Formatting] Whole numbers should be formatted with thousands separators and no decimals (7 o...		2
+ [Formatting] Relationship columns should be of integer data type (2 objects)		1
+ [Formatting] Add data category for columns (5 objects)		1
+ [Formatting] Hide foreign keys (3 objects)		2
+ [Formatting] Mark primary keys (3 objects)		1
+ [Formatting] Month (as a string) must be sorted (2 objects)		2

123 objects in violation of 16 Best Practice rules.

BPA – Manage Rules

DAX Expressions

Rules in collection:

Rule name	Scope	Severity
☐ DAX Expressions		
☒ [DAX Expressions] Avoid using '1-{x/y}' syntax	Measures, Calculated Columns, Calculation Items	2
☒ [DAX Expressions] Avoid using the IFERROR function	Measures, Calculated Columns	2
☒ [DAX Expressions] Column references should be fully qualified	Measures, KPIs, Table Permissions, Calculation Items	3
☒ [DAX Expressions] Filter column values with proper syntax	Measures, Calculated Columns, Calculation Items	2
☒ [DAX Expressions] Filter measure values by columns, not tables	Measures, Calculated Columns, Calculation Items	2
☒ [DAX Expressions] Inactive relationships that are never activated	Relationships	2
☒ [DAX Expressions] Measure references should be unqualified	Measures, Calculated Columns, Calculated Tables, KPIs, Calculation Items	3
☒ [DAX Expressions] Measures should not be direct references of other measures	Measures	2
☒ [DAX Expressions] No two measures should have the same definition	Measures	2
☒ [DAX Expressions] The EVALUATEANDLOG function should not be used in production models	Measures	1
☒ [DAX Expressions] Use the DIVIDE function for division	Measures, Calculated Columns, Calculation Items	2
☒ [DAX Expressions] Use the TREATAS function instead of INTERSECT for virtual relationships	Measures, Calculation Items	2

What about that slow pattern?



Filter by the Whole Table

```
CALCULATE (  
    [Sales Amount],  
    FILTER (Sales, Sales[Quantity] > 1  
)  
)
```

Edit Rule

Edit Best Practice Rule

Name

[DAX Expressions] Filter measure values by columns, not tables

ID

FILTER_MEASURE_VALUES_BY_COLUMNS

Severity

2

Category

DAX Expressions

Description

Option 1: FILTER(VALUES('Table'[Column]),[Measure] > Value)
Option 2: FILTER(ALL('Table'[Column]),[Measure] > Value)
Reference: <https://docs.microsoft.com/power-bi/guidance/dax-avoid-avoid-filter-as-filter-argument>

Applies to

Calculated Columns, Calculation Items, Measures

satisfying the following criteria:

Rule Expression Editor

Regex.IsMatch(Expression,"(?i)CALCULATE\s*\(\s*\^[,]+\s*(?i)FILTER\s*\(\s*\^[A-Za-z0-9 _]+\s*\s*,\s*\^[^\s]+\s*\)")
or
Regex.IsMatch(Expression,"(?i)CALCULATETABLE\s*\(\s*\^[,]*\s*(?i)FILTER\s*\(\s*\^[A-Za-z0-9 _]+\s*\s*,\s*\^[^\s]+\s*\)")

Minimum Compatibility Level

CL 1200 (SQL Server 2016 / Azure AS)

OK

Cancel

Description with help link

- Instead of using this pattern `FILTER('Table','Table'[Column]="Value")` for the filter parameters of a `CALCULATE` or `CALCULATETABLE` function, use one of the options below. As far as whether to use the `KEEPFILTERS` function, see the second reference link below.
- Option 1: `KEEPFILTERS('Table'[Column]="Value")`
- Option 2: `'Table'[Column]="Value"`
- Reference: <https://docs.microsoft.com/power-bi/guidance/dax-avoid-avoid-filter-as-filter-argument>
- Reference: <https://www.sqlbi.com/articles/using-keepfilters-in-dax/>

Another Source

9. Filtering a column

The [FILTER](#) function is often overused. Its main purpose is for filtering columns based on measure values. If you're just filtering a column value, there's generally no need to use this function. In fact, using it in that scenario often degrades performance.

Don't use this logic:

```
US Revenue = CALCULATE ( [Revenue], FILTER ( 'Geography', 'Geography'[Area] = "United States" ) )
```

Option 1:

```
US Revenue 1 = CALCULATE ( [Revenue], 'Geography'[Area] = "United States" )
```

Option 2:

```
US Revenue 2 = CALCULATE ( [Revenue], KEEPFILTERS ( 'Geography'[Area] = "United States" ) )
```

[Top 10 Power BI mistakes and their best practice solutions](https://www.elegantbi.com/post/top10bestpractices)
(<https://www.elegantbi.com/post/top10bestpractices>)

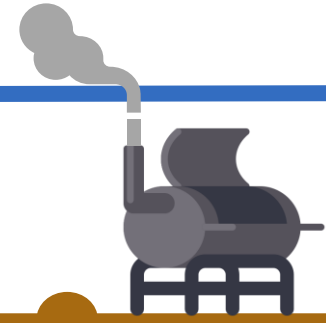
Change Description

- Link to company best practices
- Link to SharePoint that has more information

Power BI Demo

Tabular Editor

Our Journey



1. Intro
2. Performance Analyzer
3. DAX Studio
4. Tabular Editor
- 5. Conclusion**

Conclusion

- Slow Report ?
 - Start in Power BI Desktop
 - Performance Analyzer
 - DAX Studio
- Prevention
 - Tabular Editor

Take Away

- Time Invested in these tools
 - Can only enhance your skillset
 - Can make you more productive
- Power BI Desktop can only do so much

Resources



[Tabular Editor – Wonderful Training Free](#)



[Tabular Editor – Blog Posts](#)



[DAX Studio](#)



[Tabular Editor 2.x](#)

Resources



[Data Goblins - Sample Datasets](#)



[Data Mozart - Lots of DP-600 Resources](#) [Visual Speed](#)

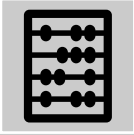


[The Definitive Guide to DAX - 2nd Edition](#)



[Optimizing DAX Book 2nd Edition](#)

Resources



[Elegant BI Blog \(Excellent Source\)](#)



[Microsoft Best Practice Rules Blog](#)



[Microsoft Github Best Practice Rules](#)

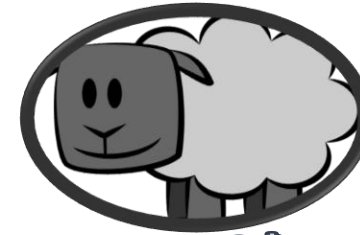


[DP-600 Exam Prep Book](#)

Resources

- Attributions:
 - Smoker icon: [Flaticon.com](https://www.flaticon.com). Icons using resources from [Flaticon.com](https://www.flaticon.com)

Thank you



The Dax Shepherd

Jason Romans
thedaxshepherd@gmail.com
www.thedaxshepherd.com

